MILITARY SPECIFICATION

CARTRIDGE, IMPULSE, MARK 8 MOD 0

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the manufacture and acceptance of the Mark 8 Mod 0 Impulse Cartridge (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Military

MIL-P-116	Preservation-Packaging, Methods of
MIL-D-21625	Design and Evaluation of Cartridges for Cartridge Actuated Devices
MIL-I-81530	Ignition Element, Electric, Mark 11 Mod 0

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-453	Inspection, Radiographic

Beneficial comments (recommendations, additions , deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 1377

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2.1.2 Drawings and publications. The following drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

Naval Air Systems Command (Code Ident 30003)

2240770	Container, Inner, for Cartridge Impulse, Mark 8 Type, Assembly
2299605	Container, Outer, for Cartridge Impulse, Mark 8 Type, Assembly
DL537909	Cartridge, Impulse, Mark 8 Mod 0
DL546659	Ignition Element, Electric, Mark 11 Type

PUBLICATIONS

Naval Sea Systems Command (Code Ident 53711)

OP 5	Ammunition and Explosives Ashore
WR-43	Preparation of Quality Assurance Provisions

Code of Federal Regulations

49 CFR 100-199	Transportation

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

2.2 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Conformance to documents. The Mark 8 Mod 0 impulse cartridge covered by this specification shall be manufactured in accordance with DL537909 and all documents listed thereon.

3.3 Materials. All materials used in the manufacture of the cartridges shall conform strictly to the specifications referred to on the respective drawings. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.4 Primary components. For the purposes of this specification the Mark 11 Mod 0 electric ignition element (DL546659) and the cartridge propellant charge are considered primary components.

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3.5 Ignition element. The ignition elements shall be furnished from lots which were manufactured within the past 12 months. The ignition elements shall be manufactured in accordance with the drawings listed on DL546659, and shall be from lots which were accepted in accordance with the requirements of MIL-I-81530 (see 4.6.1). When the same contractor manufactures the Mk 11 Mod 0 Ignition Element and the Mk 8 Impulse Cartridge and assembles the ignition element in the cartridge, the preproduction and production test requirements of MIL-I-81530 for the ignition element shall be waived.

3.6 Propellant charge.

3.6.1 Quantity. All propellant charges shall be as specified on the applicable drawing. Charges shall be determined by weight rather than by volume measurement.

3.6.2 Smokeless powder, handling, and loading. The smokeless powder shall be subjected to no unnecessary handling or exposure to the atmosphere. The handling and loading of smokeless powder exposed to the atmosphere shall be performed only under conditions that do not permit the relative humidity to exceed 75 percent.

3.6.3 Smokeless powder, condition. The smokeless powder used shall exhibit no evidence of decomposition (see 4.6.2)

3.7 Cartridge requirements.

3.7.1 Bridge circuit resistance. The resistance of the bridge circuit in each completely assembled cartridge, when measured electrically (see 4.6.4), shall be not less than 0.08 ohm nor more than 0.15 ohm.

3.7.2 Leakage. Cartridge leakage rate shall be not greater than 10^{-5} cubic centimeters of air per second (see 4.6.5).

3.7.3 X-ray. Cartridges shall be free of imperfections when X-rayed (see 4.6.6).

3.8 Ballistic requirements. Cartridges shall meet the ballistic requirements specified in 3.8.1 through 3.8.4 when subjected to the test prescribed in 4.6.7.

3.8.1 Velocity at ejection. The lower limit (LL) of velocity shall be 16.0 feet per second. The upper limit (UL) of velocity shall be 25.0 feet per second.

3.8.2 Pressure. The lower limit (LL) of pressure shall be 5,200 psi. The upper limit (UL) of pressure shall be 8,700 psi.

3.8.3 Ignition time. The time from the application of firing current to the first indication of pressure rise shall be not greater than 0.050 second.

3.8.4 Release time. The time from application of firing current to the first indication of hook opening shall be not greater than 0.020 second.

3.9 Change in propellant lot. Whenever a change occurs in the propellant lot for the main charge, test firings shall be conducted in accordance with 4.8.

3.10 Workmanship. Cartridges shall be free of the following defects: burrs, dents, incomplete or damaged threads, missing or damaged O-ring, perforated closure disc, or other defects which may prevent entry of the cartridge into the firing chamber, or affect the resistance of the cartridge to moisture (see 4.6.3).

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4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 **Responsibility for compliance.** All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 **Classification of inspections.** The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5).

4.3 **Inspection conditions.** Unless otherwise specified (see 6.2), all inspections shall be performed as specified in the applicable test method of 4.6.

4.4 **First article inspection.** First article inspection shall be performed after award of contract and prior to production by a facility designated in the contract or order (see 6.2). A first article sample acceptable for environmental and function testing shall be a sample that has met the requirements of 3.2 through 3.7.3 and 3.10 of this specification. First article inspection shall be performed on sample units which have been produced using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing shall be the same as used for regular production and shall be from the intended sources of supply.

4.4.1 **Sample size.** The first article sample shall consist of 76 cartridges.

4.4.2 **Inspection routine.** Sixty-six cartridges shall be expended in the tests listed below and 10 shall be retained for investigative purposes. Testing shall consist of the following tests as prescribed in MIL-D-21625 (see 6.2): 15 "g" shock (6); Temperature/humidity cycling [THC(18)]; vibration (6). The number in parentheses denotes the number of cartridges allocated for each test. After exposure to the specified environmental treatment, the cartridges shall be conditioned for 12 hours at 70° ± 5°F and fired, together with 12 unconditioned cartridges, as specified in 4.6.7. In addition 12 cartridges shall be conditioned at each of the following temperatures: −65°F ± 5°F and 160° ± 5°F. These cartridges shall be fired as specified in 4.7.7 to check temperature sensitivity. No cartridge shall produce results outside the limits of 3.8

4.4.3 **Failure.** Failure of any cartridge shall be cause for rejection of the first article sample.

4.5 **Quality conformance inspection.** Quality conformance inspection shall consist of the tests specified in table I.

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4.5.1 Lot size. Unless otherwise specified (see 6.2), the production lot size shall be not less than 1301 and not greater than 3200. Cartridges required for all test purposes are selected from the production lot and shall not be applied as a part of the quantity specified for delivery by the contract. Only primary components from a single lot shall be used in a production lot of cartridges; however, one primary component production lot may be used in more than one cartridge production lot. Cartridges selected for the ballistic test shall be delivered to the activity designated by the contracting agency.

4.5.2 Sampling. Unless otherwise specified (see 6.2), sampling shall be in accordance with table II.

4.5.3 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as critical, major, or minor in accordance with WR-43. Tests and examinations that verify critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of the test or examination. Tests and examinations which are not annotated with a classification code are classified minor.

TABLE I. Quality conformance tests and examinations.

Item	Nature of test	Requirement para.	Test para.	Classification (WR-43)
Primary component	Single lot verification	3.4	4.5.1	(M101)
Ignition element	Age and acceptance verification	3.5	4.6.1	(C1)
Propellant charge	Type and weight	3.6.1	-	(C2)
Smokeless powder	Condition	3.6.3	4.6.2	(C3)
Cartridge	Visual examination	3.7.4	4.6.3	(M102)
Cartridge	Bridge circuit resistance	3.7.1	4.6.4	(C4)
Cartridge	X-ray	3.7.3	4.6.6	(C5)
Cartridge	Ballistics	3.8	4.6.7	(C6)
Cartridge	Leakage	3.7.2	4.6.5	(C7)
Cartridge	Ballistics (change in propellant lot)	3.9	4.8	(C8)
Cartridge	Preservation-packaging, packing and marking	Section 5	4.6.3	(M103)

TABLE II. Quality conformance sample size.

Lot size	Sample size	Retained sample
66 - 110	15	10
111 - 180	15	10
181 - 300	20	10
301 - 500	25	10
501 - 800	30	10
801 - 1300	35	10
1301 - 3200	40	10
3201 - 8000	50	10
8001 - 22000	75	10

4.6 Tests.

4.6.1 Ignition element verification. Verification shall be provided that the ignition elements used were certified and accepted in accordance with the requirements of 3.5.

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4.6.2 Smokeless powder, condition. Smokeless powder shall be inspected in accordance with the applicable paragraphs of OP 5. Evidence of decomposition shall be cause for rejection.

4.6.3 Visual examination. All cartridges shall be visually examined. Cartridges having any of the defects of 3.10 shall be cause for rejection. The preservation-packaging, packing, and marking of the cartridges shall be examined for conformance to section 5.

4.6.4 Bridge circuit resistance. Conformance with the bridge circuit resistance of 3.7.1 shall be ascertained on all completely assembled cartridges by means of a test circuit which limits the bridge circuit current to not greater than 25 milliamperes. A safety chamber with an interlock switch shall be used to protect the operator during the application of current. A suitable instrument which is known to be accurate to within 0.005 ohm in the range of resistance specified shall be used to determine whether or not the requirement is being met. Units which fail to meet the requirements of 3.7.1 shall be cause for rejection. The results of bridge circuit resistance measurements shall not be affected by resistance of the test circuit and connector.

4.6.5 Leakage. Each cartridge in the first article or production sample shall be leak tested in a dry gas leak tester. Cartridges which exhibit a leak rate greater than 10^{-5} cc per second of air at a pressure differential of 1 ± 0.1 atmosphere shall be considered defective. Cartridges of the first article sample which exhibit a leakage defect shall be cause for rejection. If the number of cartridges of the production lot sample exhibiting leak defects exceeds an AQL of 2.5 percent defective from table II-A of MIL-STD-105, then a 100 percent leak test of the lot being tested shall be required. Cartridges which exhibit a leak rate greater than that specified above shall be rejected (see 6.5).

4.6.6 Radiographic examination. All cartridges, in the first article and production test samples, shall be radiographically examined in accordance with MIL-STD-453. Cartridges having any observable imperfections in assembly shall be cause for rejection. The cartridges shall be positioned on their sides for the most revealing exposure. All cartridges shall be identified with serial numbers prior to examination. The cartridge serial numbers shall be in consecutive order beginning with the number 001 in each production lot. The cartridges shall be arranged on trays or boards in consecutive numerical order, and each radiograph shall carry a permanent identification of the cartridges displayed thereon. The radiograph identification shall include the complete lot number, as stamped on the cartridges, and the span of serial numbers displayed. If any assembly or loading defects are noted in the production lot samples, the contractor shall X-ray the represented lot 100 percent and make available the X-rays to the cartridge design activity for review. Discontinuities in serial numbers shall be identified on the X-ray. Defective cartridges found by the vendor shall be clearly identified on the X-ray and those defectives removed from the production lot. The serial numbers shall be pen or ink stamped on the cartridges with approximately 1/8-inch-high characters. The serial numbers shall be located apart from the other cartridge markings.

4.6.7 Ballistic test. The number of cartridges selected from each production lot for the ballistic test shall be in accordance with table II. For test firing the Mark 8 impulse cartridges, a 1000-pound general purpose inertly loaded store shall be mounted on a standard Aero 7/A bomb ejector rack. The position of the ejector shall be horizontal. Suitable instrumentation for measuring chamber pressure and for determining the time from the closing of the firing circuit to the complete opening of one of the ejector hooks shall be installed. The cartridges shall be conditioned for 12 hours at 70° ± 5°F prior to firing in the ejector rack. Firing current shall be 10 ± 0.5 amperes dc. Any cartridge which fails the requirements stated in 3.8 shall be cause for rejection of the lot represented.

4.6.7.1 Velocity at ejection. The release velocity shall be measured 6 inches from the end of the ejection stroke.

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4.7 **Retest.** There shall be no retests.

4.7.1 **Test failure.** If test failure is attributable to an assignable cause, excluding the test cartridge, original test results should be discarded and that portion of the test reconducted.

4.8 **Change in propellant lot.** Whenever a change in propellant lot for the main charge occurs, the charge weight shall be verified by firing 36 cartridges which have been temperature conditioned as follows: 12 at $-65° \pm 5°F$, 12 at $70° \pm 5°F$, and 12 at $160° \pm 5°F$. The temperature conditioning and firing of these cartridges and the firing results shall be the same as the cartridge allocated for the low, high, and normal temperature firing tests specified in 4.4.2.

5. PACKAGING

5.1 **Preservation and packaging.** For definitions of packaging levels, see 6.4.

5.1.1 **Level A.** Level A packaging shall be used for packaging of all production lots for service use. Four cartridges shall be packaged in an inner container in accordance with Drawing 2240770.

5.1.2 **Level B.** Not applicable.

5.1.3 **Level C.** When specified in the contract (see 6.2), those cartridges which are intended for expenditure in first article sample testing and production lot acceptance testing shall be packaged in accordance with method III of MIL-P-116 and with 49 CFR 171-178.

5.2 **Packing.** For definitions of packing levels, see 6.4.

5.2.1 **Level A.** Level A packing shall be used for packing of all production lots for service use. Twelve inner containers as described in 5.1.1 shall be packed in accordance with Drawing 2299605.

5.2.2 **Level B.** Not applicable.

5.2.3 **Level C.** When specified in the contract (see 6.2), cartridges intended for expenditure in first article testing and for production lot acceptance tests and are packaged in accordance with 5.1.3 shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Shipping containers shall be in accordance with 49 CFR 171-178.

5.3 **Marking.**

5.3.1 **Special marking.** Marking of exterior containers shall be in accordance with 49 CFR 171-178.

5.3.2 **Normal marking.** Unless otherwise specified in the contract or order, the marking information on unit packages and shipping containers shall be as specified on Drawings 2240770 and 2299605, respectively. The specified marking information shall be applied to the containers in accordance with the applicable provisions of MIL-STD-129.

6. NOTES

6.1 **Intended use.** The Mark 8 Mod 0 impulse cartridge is intended for use in Navy bomb ejection racks.

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6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this specification
b. Whether first article inspection is required, and if so, specify the test activity
c. Date and place of delivery
d. Inspection conditions, if other than as specified (see 4.3)
e. That revision "D" of MIL-D-21625 be used (see 4.4.2)
f. Lot size, if other than as specified (see 4.5.1)
g. Sampling, if other than as specified (see 4.5.2)
h. Special marking, if any (see 5.3)
i. That the safety precaution requirements of the "Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material," DoD 4145.26M are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore," OP5 are applicable.

6.3 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of 76 cartridges. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Definitions. For the purpose of this specification, the following definitions apply.

6.4.1 Level A packaging. The degree of preservation-packaging which will afford adequate protection against corrosion, deterioration and physical damage during handling, shipment, indeterminate storage and world-wide redistribution.

6.4.2 Level C packaging. The degree of preservation-packaging which will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity for immediate use. This level may conform to the supplier's commercial practice when such meets the requirements of this level.

6.4.3 Level A packing. The degree of packing which will afford adequate protection during shipment, handling, indeterminate storage and world-wide redistribution.

6.4.4 Level C packing. The degree of packing which will afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. This level in general will conform to applicable carrier rules and regulations and may be the supplier's commercial practice when such meets the requirements of this level.

6.4.5 Hangfire. If the interval from application of current to upper hook opening time is greater than 0.020 second, the cartridge shall be considered defective and the lot considered unacceptable. Time measurement shall be recorded for each cartridge firing.

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6.5 Advisory leak test. Indian Head leak tests cartridges in accordance with "Cartridge Leak Test Procedures at the Naval Ordnance Station, Indian Head, Md."

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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